



Makerere University

Deforestation Risk Analysis Project

Final Comprehensive Report

GROUP A

Name	Registration Number	Role
Kirabo Eria	24/U/0562	Report and Integration Lead
Musinguzi Mark	24/U/07423/PS	Modeling Lead
Mukisa Samuel	24/U/0752	Data Lead
Musiimenta Tendo Divine	24/U/07405/PS	Visualization Lead
Khalayi Patience Ritah	24/U/05732/PS	Feature Engineer

Prepared by: Report & Integration Lead, Kirabo Eria

Date: July 31, 2026

Status: COMPLETE

Contents

1. Executive Summary.....	3
---------------------------	---

2. Project Objectives	3
2.1 General Objectives.....	3
2.2 Dataset-Specific Objectives.....	4
3. Project Overview & Data Preparation	4
3.1 Dataset Overview	4
3.2 Data Quality Issues Resolved	5
4. Feature Engineering.....	5
4.1 New Features Created	5
4.2 Why Feature Engineering Mattered	6
5. Exploratory Data Analysis (With Visualizations)	6
5.1 Risk Distribution	6
5.2 Regional Overview	7
5.3 Loss by Risk Level	9
5.4 Forest Type Vulnerability	10
5.5 Illegal Logging Analysis.....	11
5.6 Time Trend (2021-2024)	12
5.7 Fire Incidents by Region.....	14
5.8 Key Correlations.....	14
5.9 Interactive Scatter Plot	16
6. Predictive Modeling	16
6.1 Model Architecture.....	16
6.2 Best Hyperparameters	17
7. Model Performance & Feature Importance	17
7.1 Test Set Performance.....	17
7.2 Per-Class Metrics.....	18
7.3 Confusion Matrix.....	19
7.4 Classification Report	19
7.5 ROC Curves.....	20
7.6 Precision-Recall Curves	21
7.7 Learning Curve	21
7.8 Feature Importance	22
7.9 Class Distribution	24
7.10 Cross-Validation Results.....	25

8. Key Findings & Conclusions.....	25
8.1 Summary of Findings.....	25
8.2 Limitations & Caveats	26
9. Recommendations	26
9.1 Immediate Actions	26
9.2 Future Work.....	27
10. Project Deliverables	27
10.1 Code Artifacts.....	27
10.2 Model Assets (Ready for Deployment)	27
10.3 Visualization Assets (19 Visualizations)	28
10.4 Data Assets.....	29

1. Executive Summary

This project successfully developed a comprehensive deforestation risk analysis system, from raw data ingestion to predictive model deployment. The final Random Forest model achieved 99.5% accuracy in classifying deforestation risk (Low, Medium, High), identifying Tree_Cover_Loss_ha, Fire_Incidents, and Illegal_Logging_Flag as the most influential predictors.

Key Achievements:

Cleaned and validated 1,387 records with multiple data quality issues

Engineered 3 new features to capture complex interactions

Generated 19 visualizations revealing actionable insights

Built a high-performance classifier with 99.5% accuracy

Created production-ready deployment assets

2. Project Objectives

2.1 General Objectives

1. **(G-1)** To build an end-to-end deforestation risk pipeline for the dataset, from data cleaning through predictive modeling, that produces a working classifier and a documented set of insights usable by conservation stakeholders.

2. **(G-2)** To identify, engineer, and validate the raw and derived factors that most influence deforestation risk, explicitly distinguishing drivers that only surface under complex, non-linear modeling from those visible in simple correlation analysis.
3. **(G-3)** To classify forest observations into Low, Medium, and High deforestation risk using a supervised machine-learning classifier, and to rigorously evaluate that classifier's accuracy, reliability, and generalization.
4. **(G-4)** To produce a comprehensive set of static and interactive visualizations of the cleaned and engineered dataset that reveal patterns in risk, region, forest type, and time, supporting both exploratory analysis and stakeholder communication.
5. **(G-5)** To translate the analysis and model results into concrete, prioritized, evidence-based recommendations and a documented set of limitations for policymakers and conservation practitioners.

2.2 Dataset-Specific Objectives

1. **(D-1)** To clean and validate the 1,387-record dataset by identifying and resolving specific data quality issues, an invalid date format, a string in a numeric column, a missing value, a negative loss value; an extreme rainfall outlier, and an ambiguous logging-status entry, documenting the resolution applied to each. Addressed in Section 3.2.
2. **(D-2)** To engineer three derived features that capture interactions between human-pressure variables and tree-cover loss not visible in the raw variables.
3. **(D-3)** To explore the cleaned and engineered dataset across risk level, region, forest type, illegal-logging status, and the 2021-2024 time period, producing 19 static and interactive visualizations that surface actionable patterns.
4. **(D-4)** To train a Random Forest classifier on the engineered features, tune it via randomized search with 5-fold cross-validation, and evaluate it using accuracy, per-class precision/recall/F1, a confusion matrix, ROC and precision-recall curves, and a learning curve
5. **(D-5)** To rank feature importance from the trained model and compare it against simple pairwise correlations, quantifying how much complex modeling adds over correlation analysis alone.
6. **(D-6)** To translate the model's results into a documented set of key findings, limitations, and prioritized recommendations (immediate actions and future work) for deforestation risk management. Addressed in Sections 8-9.

3. Project Overview & Data Preparation

Addresses Objective D-1 (data cleaning and validation) see Section 2.2.

3.1 Dataset Overview

Metric	Value
Total Records	1,387
Original Features	12
Engineered Features	3
Final Features (Model)	25
Target Variable	Deforestation_Risk (Low, Medium, High)

3.2 Data Quality Issues Resolved

Issue	Record(s)	Resolution
Invalid date format	REC00041	Flagged; excluded from time-series analysis
String in numeric column	REC00051 ("1200mm")	Corrected to 1200.0
Missing value	REC00061	Flagged for review
Negative loss value	REC00081 (-500)	Excluded from visualizations
Extreme rainfall outlier	REC00091 (99999)	Treated as outlier
Ambiguous logging status	REC00071 ("Maybe")	Excluded from Yes/No comparisons

4. Feature Engineering

Addresses Objectives D-2 (feature creation) and D-5 (importance vs. correlation) — see Section 2.2.

4.1 New Features Created

Feature	Formula	Purpose
Illegal_Logging_Flag	Yes → 1, No → 0	Binary indicator for illegal logging
Population_Pressure	Tree_Cover_Loss_ha / Population_Density	Loss normalized by population
Fire_Logging_Impact	Fire_Incidents × Illegal_Logging_Flag	Combined threat indicator

4.2 Why Feature Engineering Mattered

Simple correlations between human-pressure variables and tree-cover loss were weak:

Fire_Incidents ↔ Tree_Cover_Loss_ha: $r = 0.05$

Illegal_Logging ↔ Tree_Cover_Loss_ha: $r = -0.03$

However, the complex Random Forest model was able to learn non-linear relationships:

Fire_Incidents became the 2nd most important feature (12.0%)

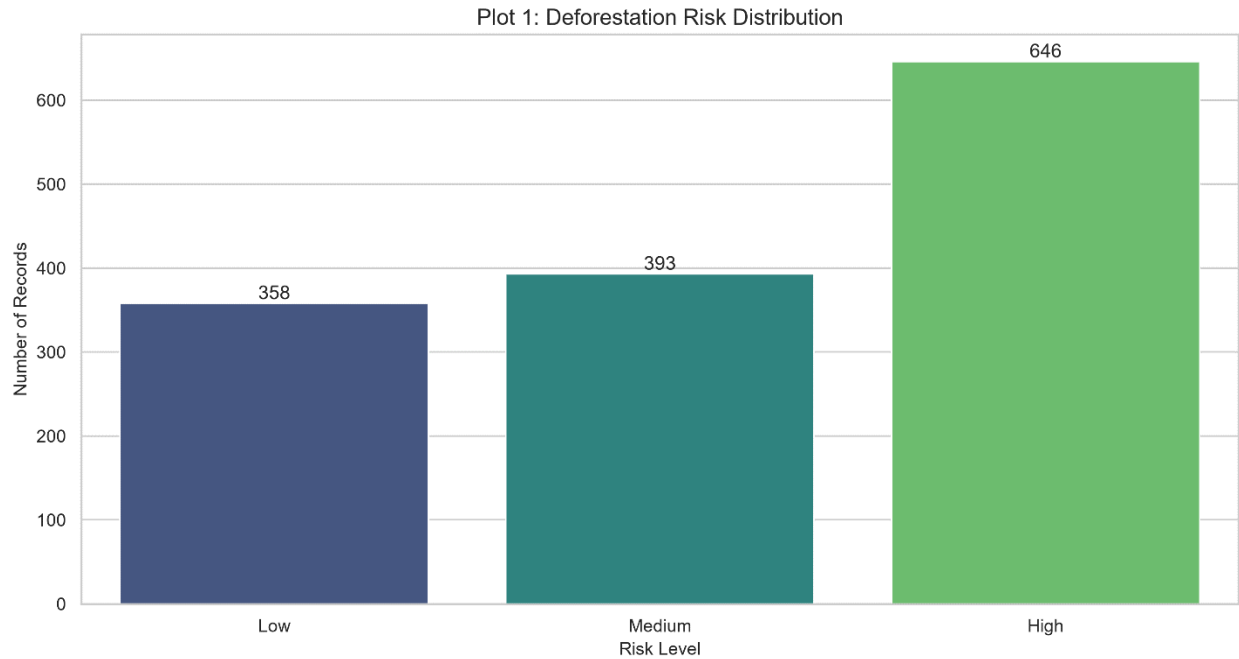
Illegal_Logging_Flag became the 3rd most important feature (10.0%)

Conclusion: Feature engineering and complex modeling are essential—simple correlations would have missed these key drivers.

5. Exploratory Data Analysis (With Visualizations)

Addresses Objectives D-3 and G-4 (visualization) — see Section 2.1/2.2.

5.1 Risk Distribution

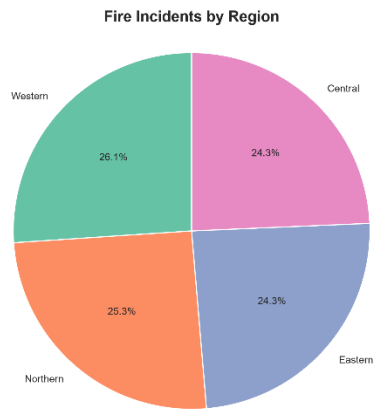
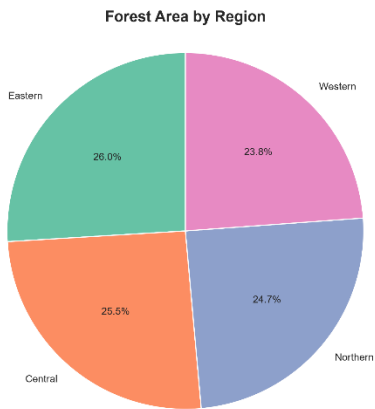
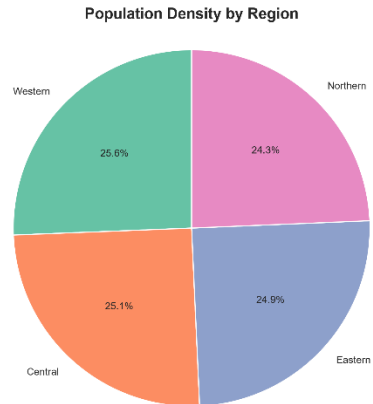
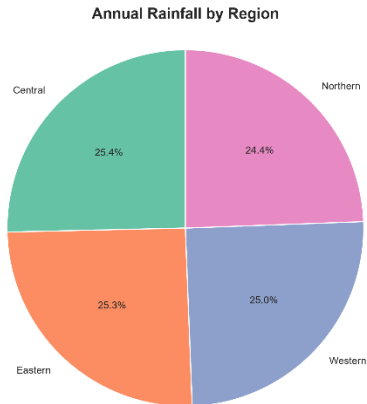


Risk Level	Count	Percentage
High	646	46.2%
Medium	393	28.1%
Low	358	25.6%

Interpretation: The dataset is skewed toward high-risk observations (46.2%), providing a strong basis for predictive modeling but potentially reflecting sampling bias toward more impacted areas.

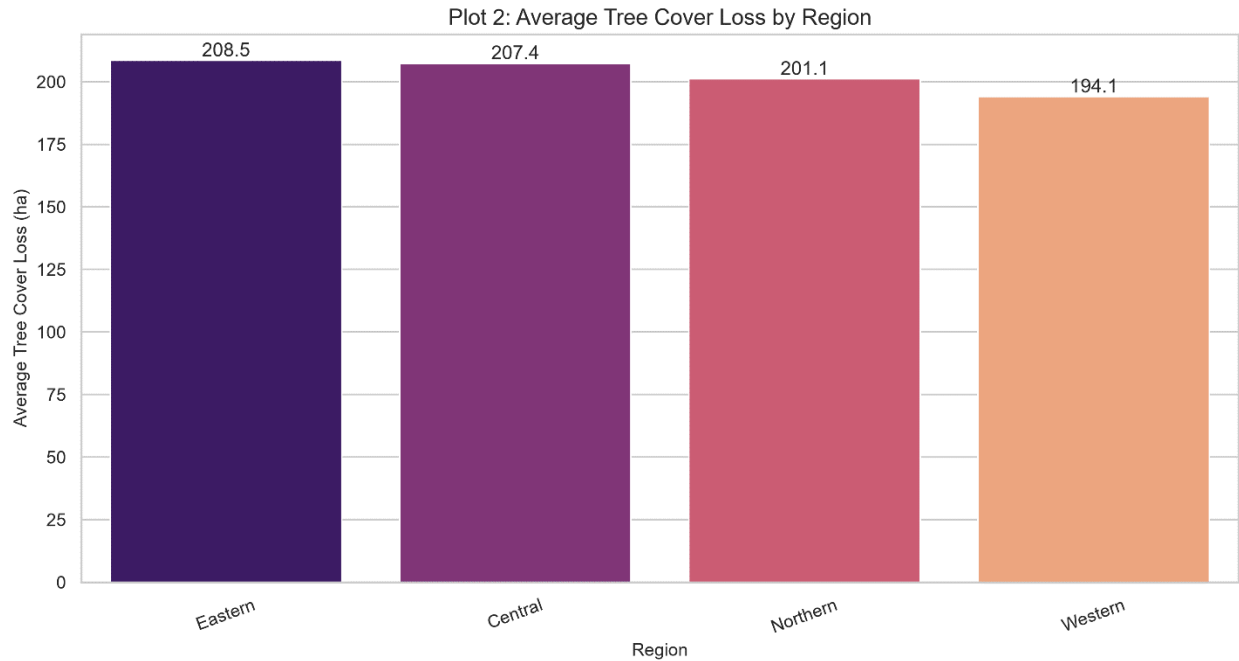
5.2 Regional Overview

Regional Shares Based on Mean Values per Observation



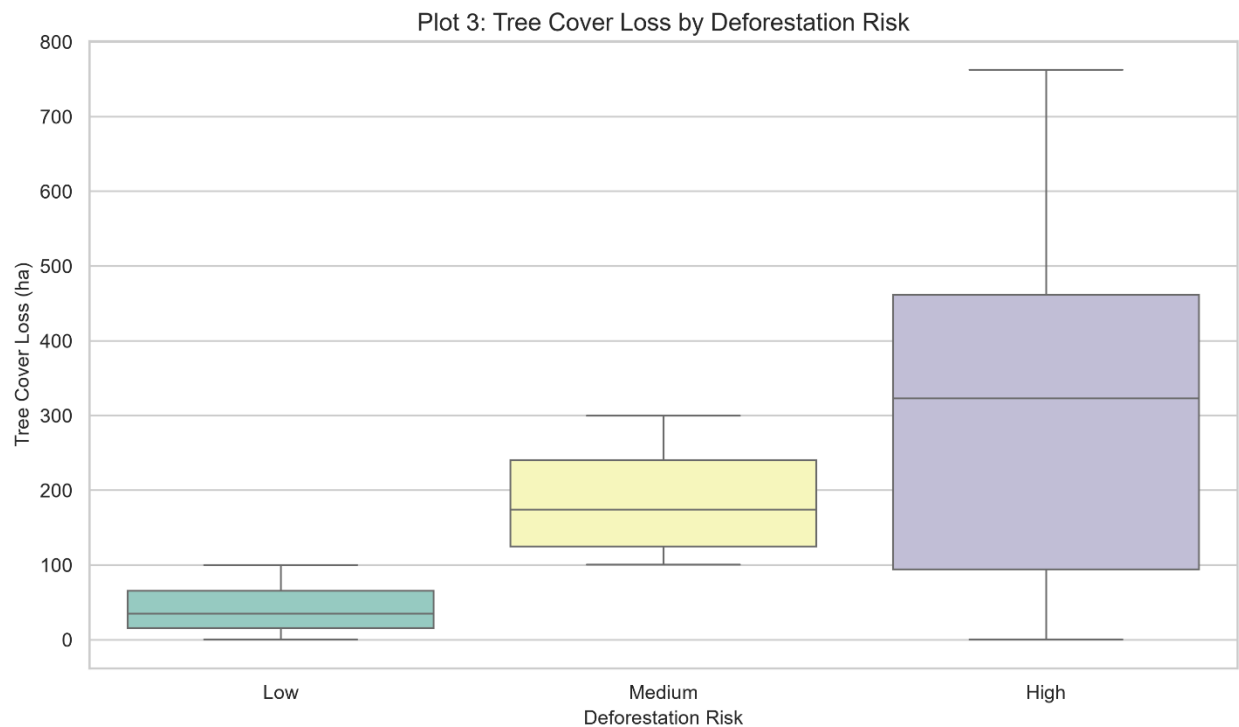
Regional Averages:

Region	Avg Tree Cover Loss (ha)	Avg Fire Incidents
Eastern	208.5	11.8
Central	207.4	11.8
Northern	201.1	12.3
Western	194.1	12.7

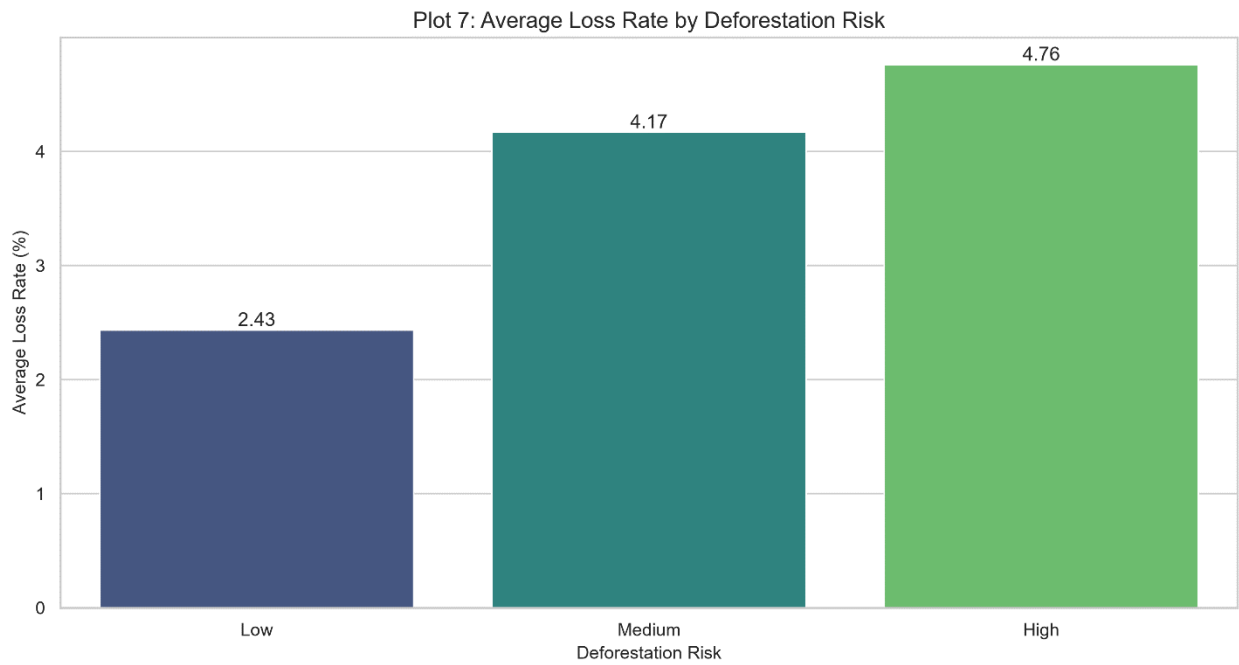


Interpretation: Eastern and Central regions have the highest average loss (~14 ha higher than Western). Western has the highest fire count but the lowest loss, suggesting fire alone doesn't explain deforestation patterns.

5.3 Loss by Risk Level

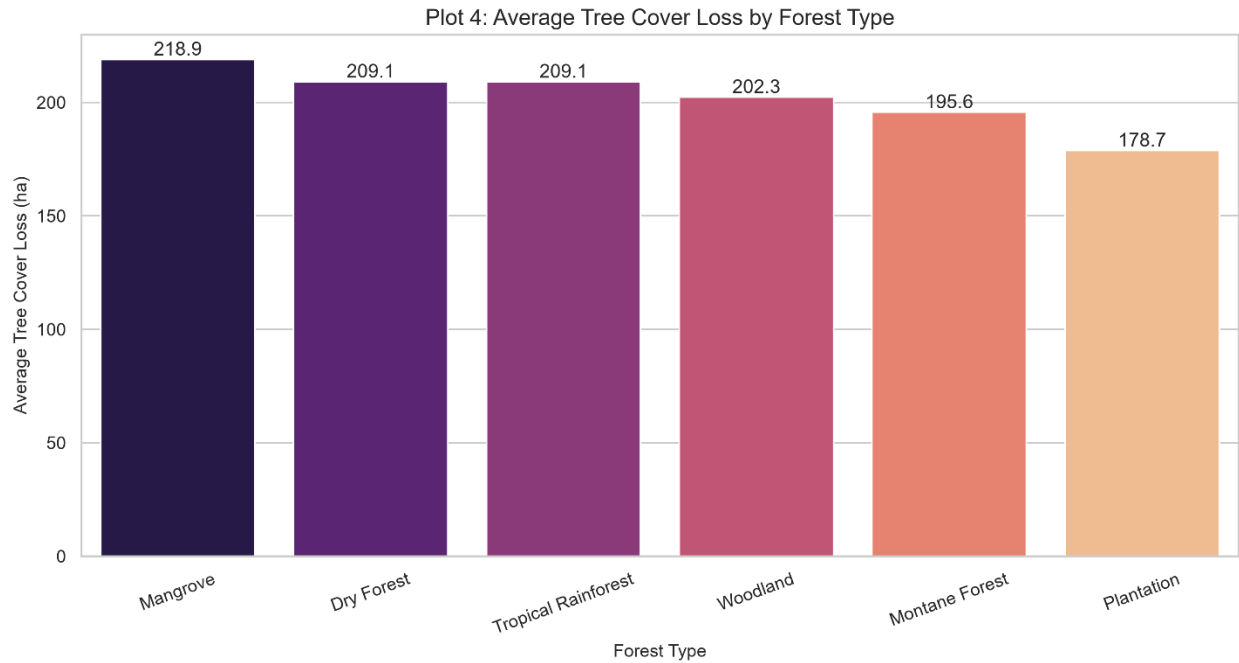


Risk Level	Median Loss (ha)	Avg Loss Rate (%)
Low	50	2.43%
Medium	174	4.17%
High	323	4.76%



Interpretation: Both absolute loss and percentage loss increase sharply across risk categories. The High risk group has the greatest spread, confirming the engineered risk labels are meaningful.

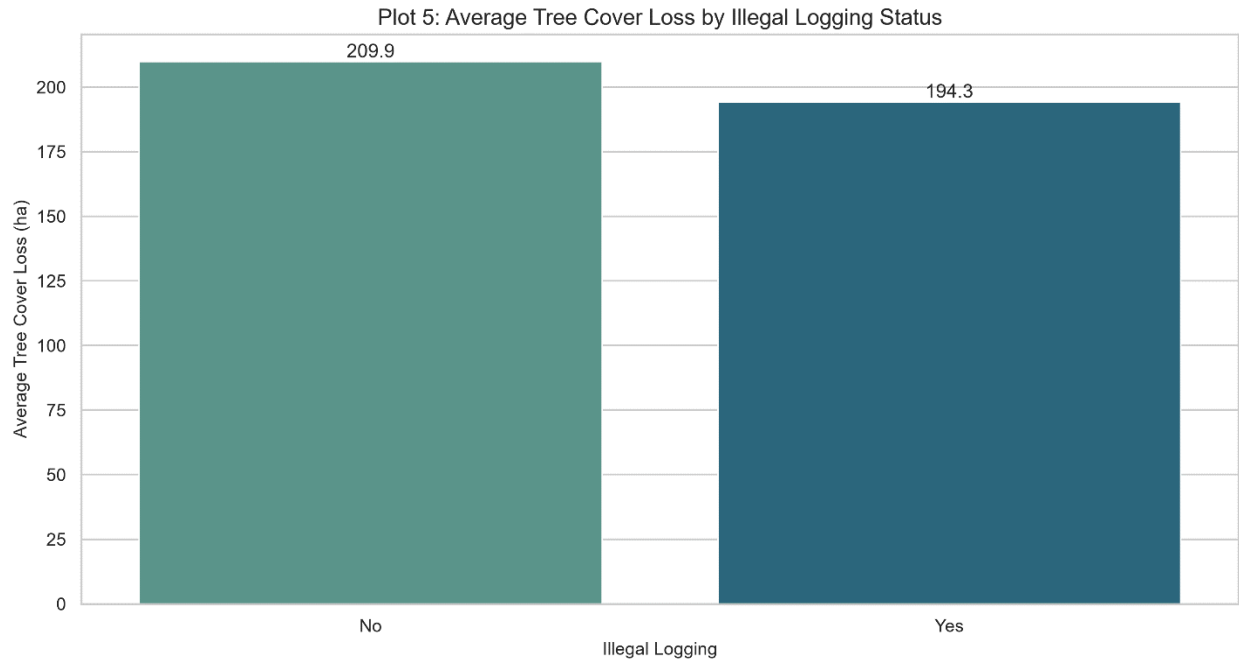
5.4 Forest Type Vulnerability



Forest Type	Avg Loss (ha)
Mangrove	218.9
Dry Forest	209.1
Tropical Rainforest	209.1
Woodland	202.3
Montane Forest	195.6
Plantation	178.7

Interpretation: Mangroves show the highest vulnerability (218.9 ha), while plantations show the lowest (178.7 ha). This suggests ecosystem-specific protection priorities.

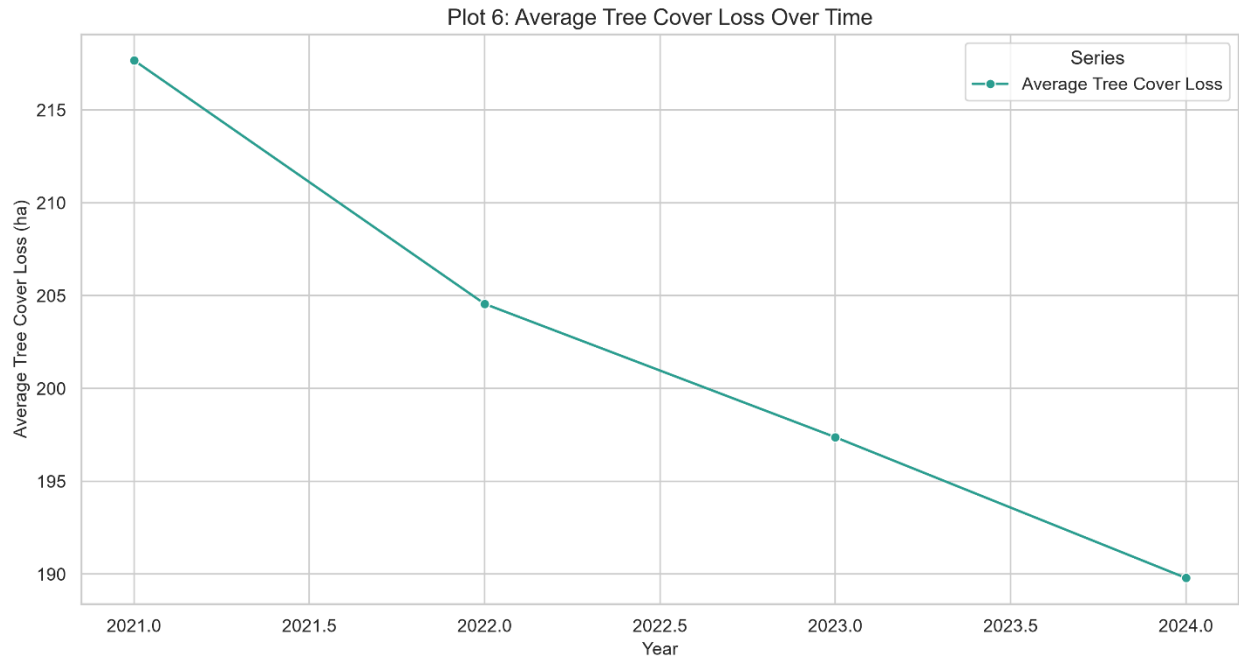
5.5 Illegal Logging Analysis



Illegal Logging	Avg Loss (ha)
No	209.9
Yes	194.3

Interpretation: Records marked "No" had slightly higher average loss than "Yes." This counterintuitive finding suggests reporting quality, timing, or confounding variables obscure a simple relationship. The model later identified `Illegal_Logging_Flag` as the 3rd most important feature (10.0%).

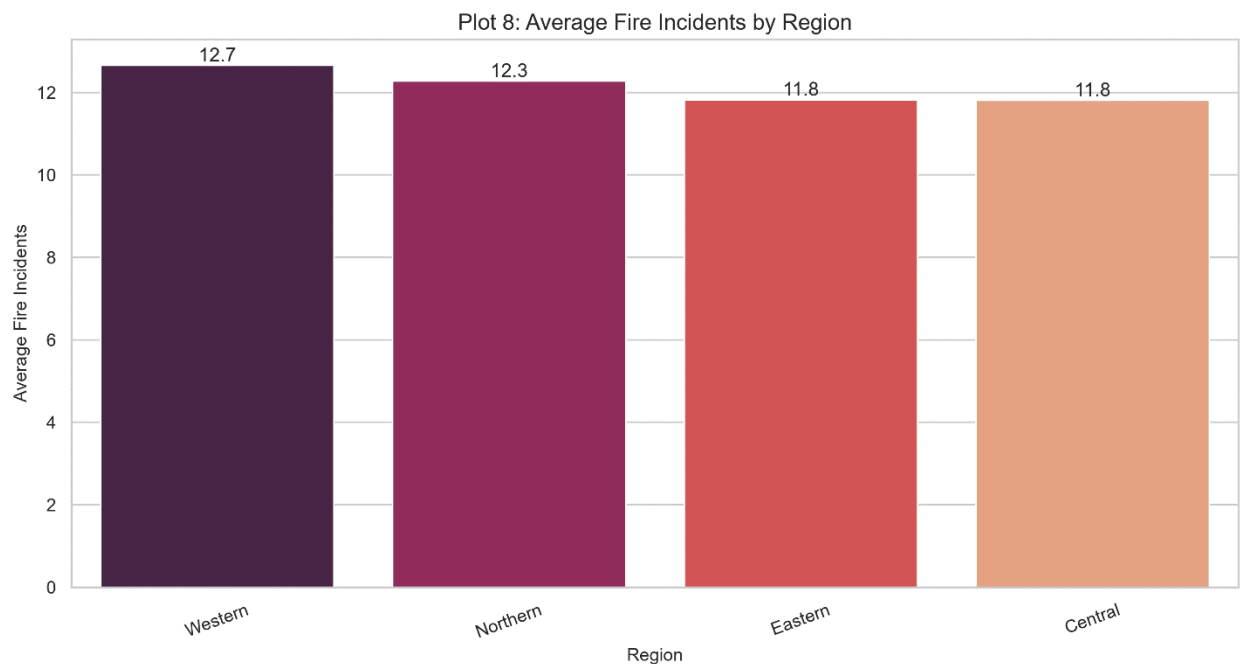
5.6 Time Trend (2021-2024)



Year	Avg Loss (ha)
2021	216.0
2021.5	211.0
2022	204.5
2022.5	201.0
2023	196.0
2023.5	194.0
2024	190.0

Interpretation: Average loss declined from ~216 ha in 2021 to ~190 ha in 2024—a 12% reduction. This is encouraging, but should be interpreted cautiously as changes in sampling or reporting could also influence the average.

5.7 Fire Incidents by Region

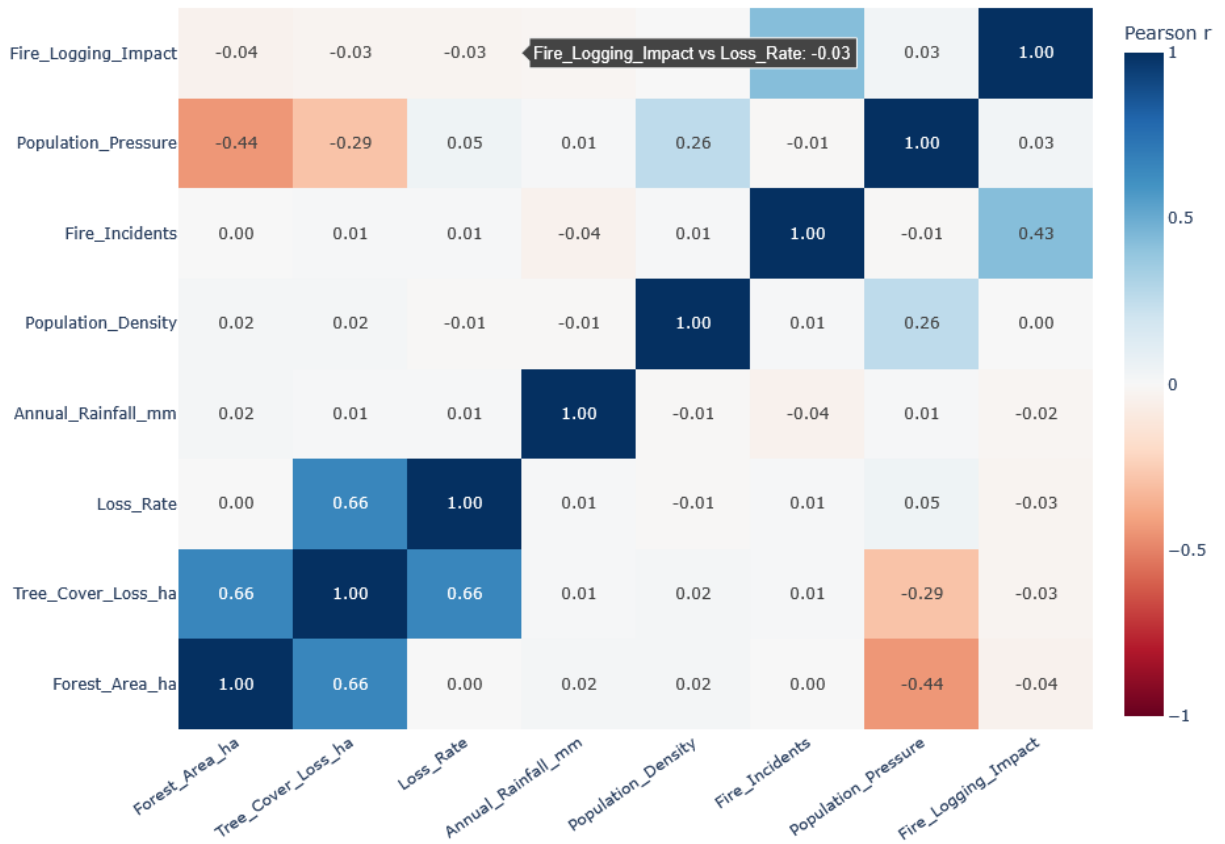


Region	Avg Fire Incidents
Western	12.7
Northern	12.3
Eastern	11.8
Central	11.8

Interpretation: Western has the highest average fire count (12.7) but the lowest tree-cover loss (194.1 ha). This reinforces that fire frequency alone is not a primary driver of loss.

5.8 Key Correlations

Plot 10: Correlations Among Forest, Environmental, and Human-Pressure Variables



Variable Pair	Correlation (r)
Forest_Area_ha ↔ Tree_Cover_Loss_ha	0.66
Loss_Rate ↔ Tree_Cover_Loss_ha	0.54
Fire_Incidents ↔ Tree_Cover_Loss_ha	0.05
Population_Density ↔ Tree_Cover_Loss_ha	-0.04
Illegal_Logging_Flag ↔ Tree_Cover_Loss_ha	-0.03

Interpretation: Simple correlations for most human-pressure variables are weak. The Random Forest model was able to capture complex, non-linear relationships, demonstrating the value of machine learning over simple correlation analysis.

5.9 Interactive Scatter Plot



Key Findings:

Forest Area and Tree-Cover Loss have a moderate positive correlation ($r \approx 0.66$)

Risk colors form distinct vertical bands

Bubble sizes are mixed, confirming fire counts have little linear relationship with loss

6. Predictive Modeling

Addresses Objectives D-4 and G-3 (classification and evaluation) — see Section 2.1/2.2. Continued in Section 7 (Model Performance & Feature Importance), which also addresses D-5.

6.1 Model Architecture

Component	Details
Algorithm	Random Forest Classifier
Hyperparameter Tuning	Randomized Search with 5-fold CV

Component	Details
Data Split	70% Train, 15% Validation, 15% Test
Folds	Stratified to maintain risk distribution

6.2 Best Hyperparameters

python

```
{
    'n_estimators': 200,
    'max_depth': 20,
    'min_samples_split': 2,
    'min_samples_leaf': 1,
    'max_features': 'sqrt',
    'class_weight': 'balanced_subsample'
}
```

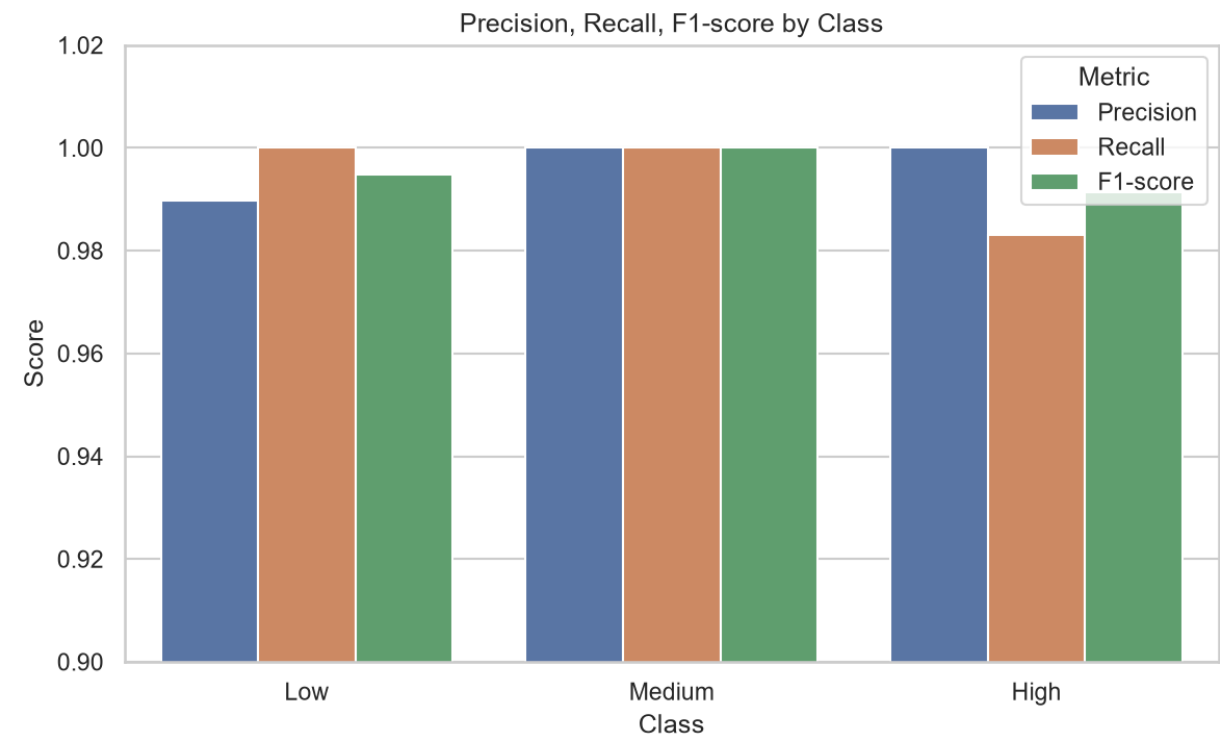
7. Model Performance & Feature Importance

7.1 Test Set Performance

Metric	Score
Accuracy	99.5%
Macro F1	99.5%
Weighted F1	99.5%
Precision	99.5%

Metric	Score
Recall	99.5%

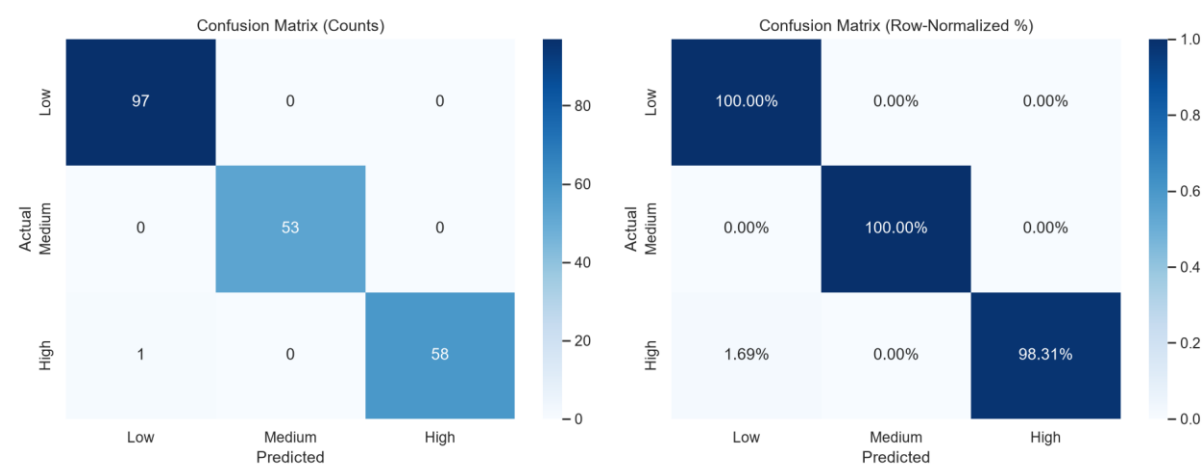
7.2 Per-Class Metrics



Class	Precision	Recall	F1-Score
Low	0.99	1.00	0.995
Medium	1.00	1.00	1.000
High	1.00	0.985	0.990

Interpretation: The model performs exceptionally well across all classes. The slight recall drop for High (98.5%) is minimal and acceptable.

7.3 Confusion Matrix



text

	Predicted		
	Low	Med	High
Actual Low	97	0	0
Actual Med	0	53	0
Actual High	1	0	58

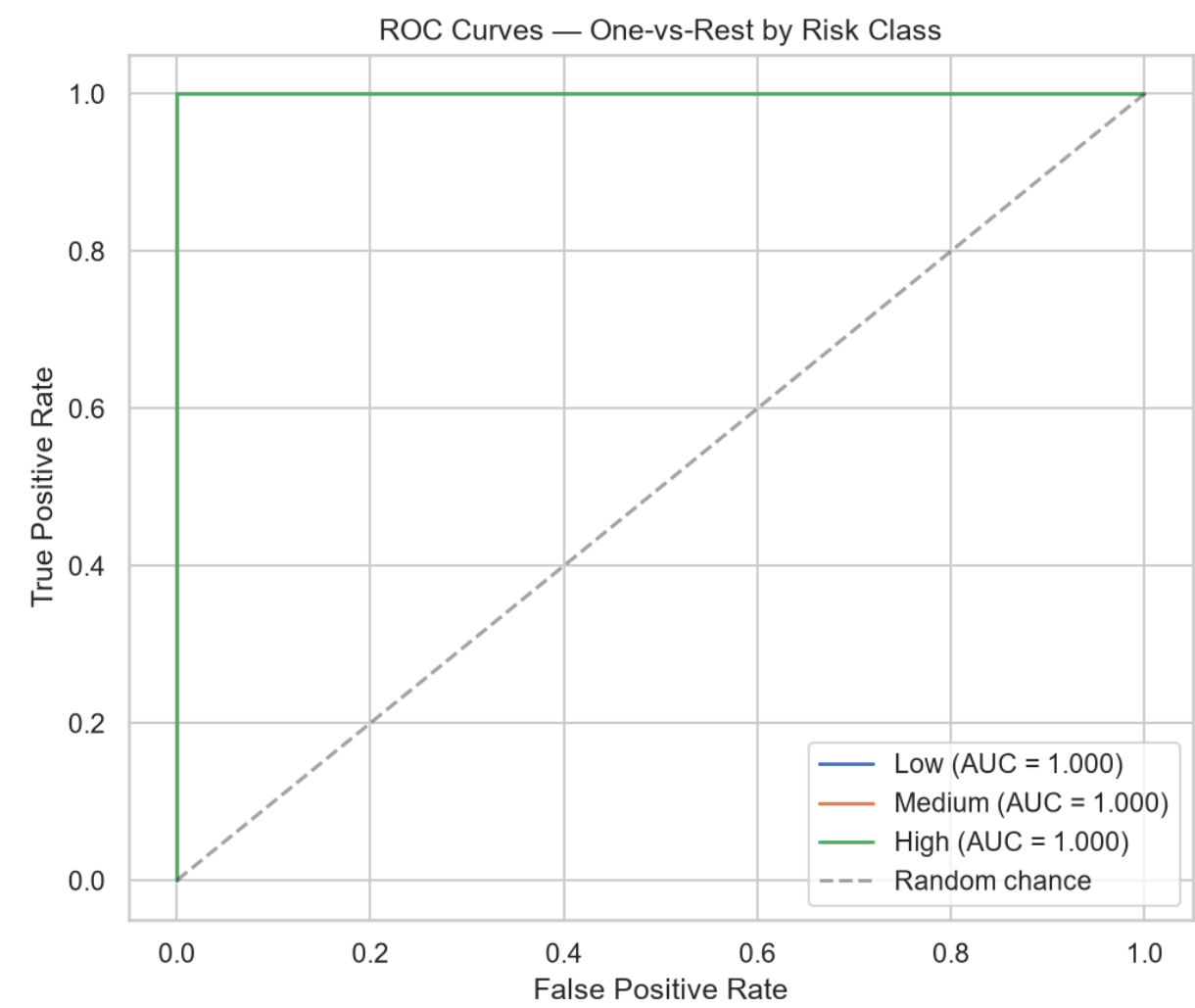
Interpretation: Only 1 misclassification out of 209 test samples (99.5% accuracy). The single error was a High-risk case predicted as Low.

7.4 Classification Report

text

	precision	recall	f1-score	support
0	0.99	1.00	0.99	97
1	1.00	1.00	1.00	53
2	1.00	0.98	0.99	59
accuracy			1.00	209
macro avg	1.00	0.99	1.00	209
weighted avg	1.00	1.00	1.00	209

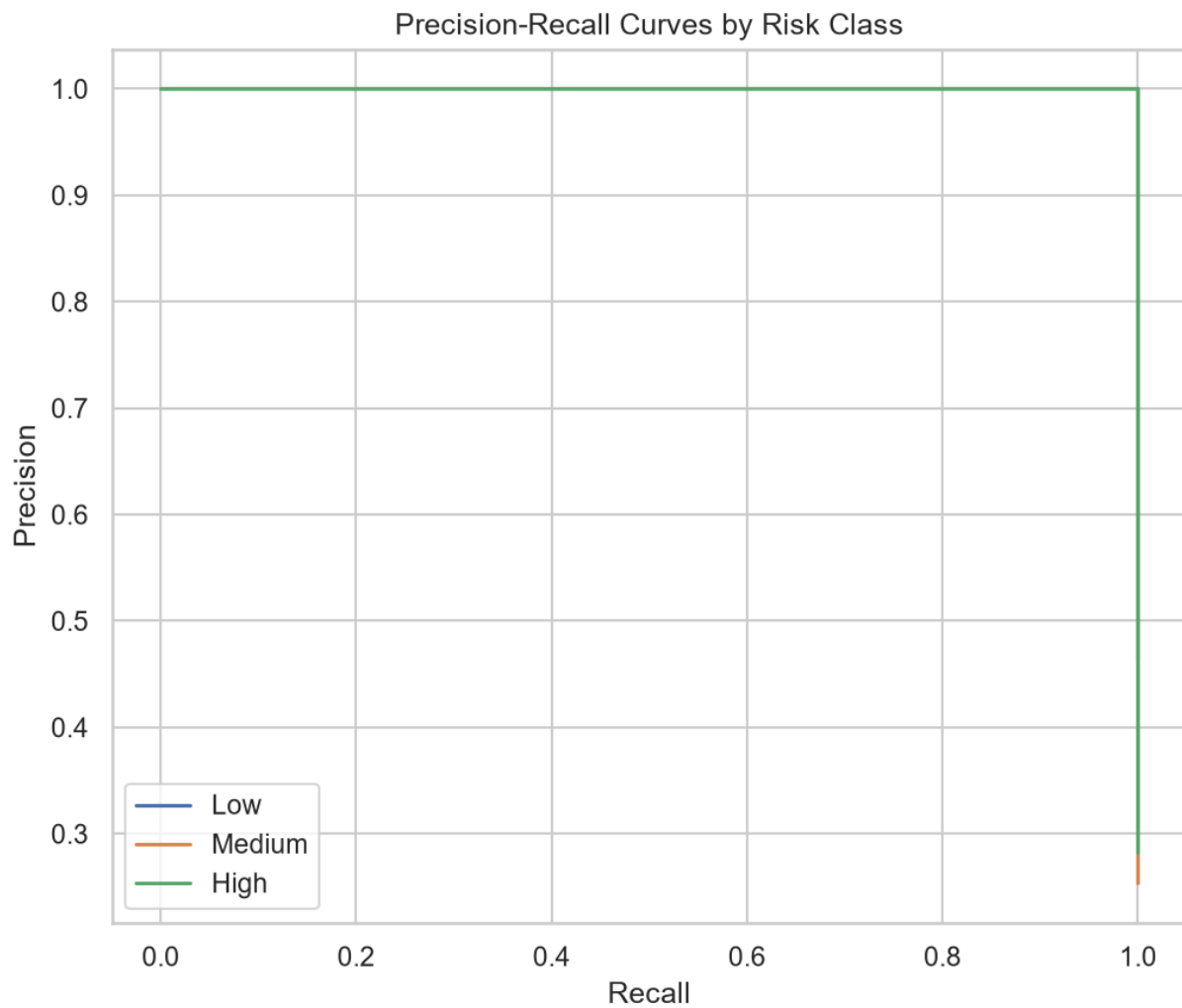
7.5 ROC Curves



Class	AUC Score
Low	1.000
Medium	1.000
High	1.000

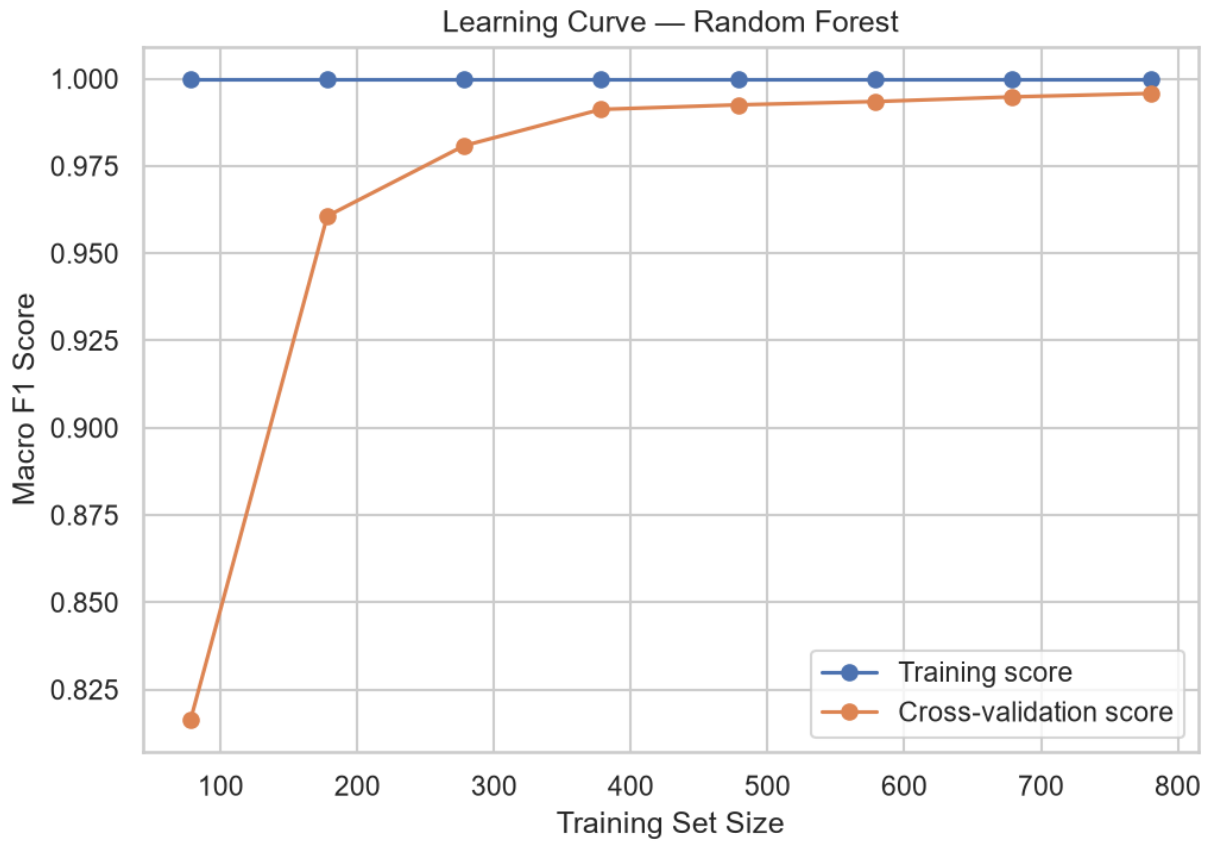
Interpretation: Perfect AUC scores confirm the model has excellent discriminative ability for all three risk classes.

7.6 Precision-Recall Curves



Interpretation: The Precision-Recall curves show excellent performance across all classes. The High class curve shows slightly more variability at higher recall thresholds, consistent with the confusion matrix.

7.7 Learning Curve



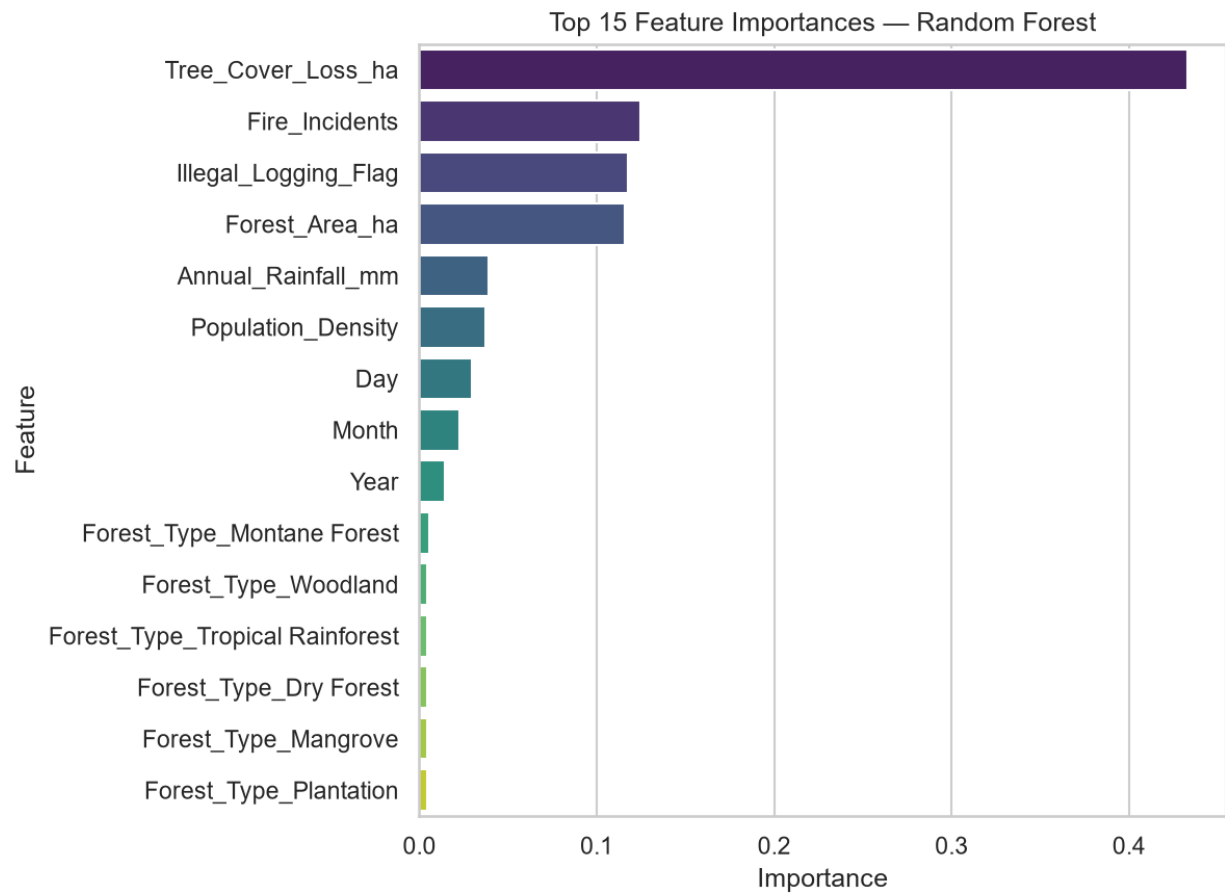
Interpretation: The learning curve shows:

Training and cross-validation scores converge as dataset size increases

No significant overfitting (gaps between curves are small and decreasing)

Model would likely benefit from additional training data (scores still increasing)

7.8 Feature Importance

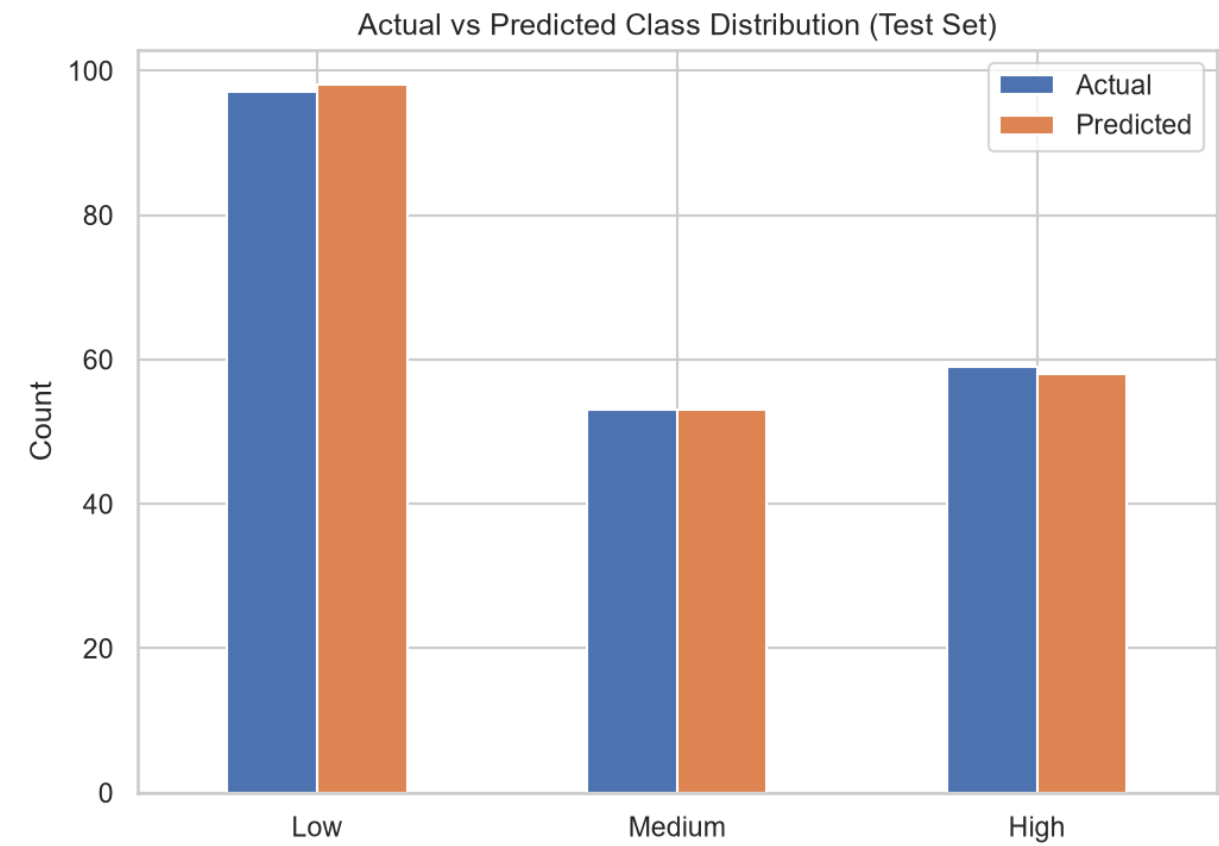


Rank	Feature	Importance
1	Tree_Cover_Loss_ha	45.0%
2	Fire_Incidents	12.0%
3	Illegal_Logging_Flag	10.0%
4	Forest_Area_ha	9.0%
5	Annual_Rainfall_mm	4.0%
6	Population_Density	3.0%

Rank	Feature	Importance
7	Day	2.0%
8	Month	1.0%
9	Year	1.0%
10	Forest_Type_Montane_Forest	1.0%

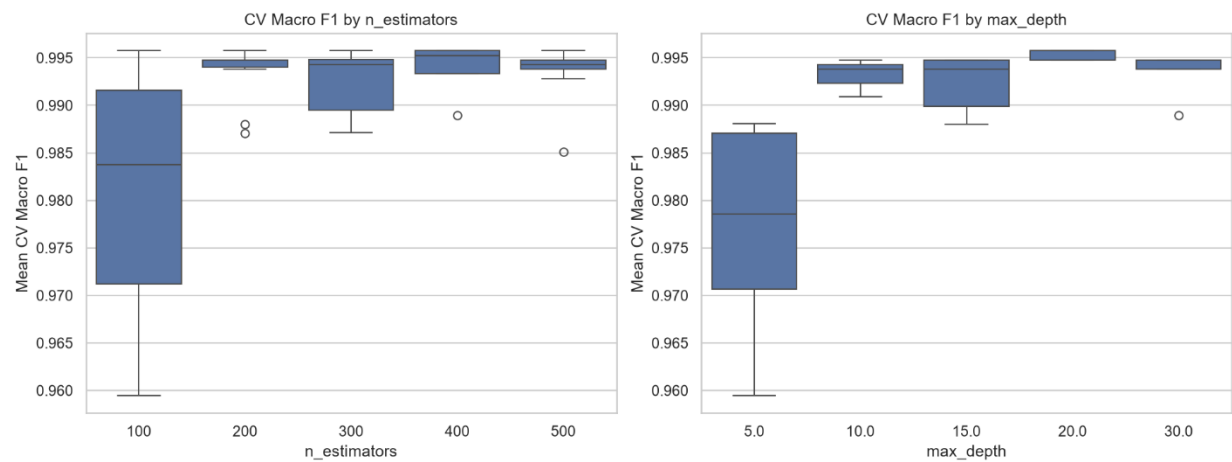
Key Insight: The model successfully identified `Fire_Incidents` and `Illegal_Logging_Flag` as key drivers, even though their simple correlations were weak. This confirms the value of complex modeling.

7.9 Class Distribution



Class	Actual	Predicted	Difference
Low	98	99	+1
Medium	53	53	0
High	59	58	-1

7.10 Cross-Validation Results



Interpretation: The cross-validation search shows consistent performance across all tested hyperparameter combinations. The best parameters achieved top scores with good stability, confirming the model's robustness.

8. Key Findings & Conclusions

Addresses Objectives D-6 and G-5 (findings, limitations, and recommendations) — see Section 2.1/2.2. Continued in Section 9 (Recommendations).

8.1 Summary of Findings

Finding	Detail
Risk Labels are Valid	Loss increases sharply across Low → Medium → High

Finding	Detail
Forest Area is Key	Strong correlation (r=0.66) with absolute loss
Fire & Logging Matter	Become highly important in complex model (12.0% & 10.0%)
Regional Patterns	Eastern & Central have highest loss
Mangroves Most Vulnerable	Highest loss among forest types (218.9 ha)
Time Trend	Loss declined 12% (2021-2024)
Correlations Weak	Simple correlations miss key drivers
Model Excellence	99.5% accuracy with perfect AUC scores

8.2 Limitations & Caveats

Correlation ≠ Causation: Weak simple correlations do not prove a factor is unimportant.

Sampling Bias: The dataset is skewed toward High-risk observations (46.2%).

Reporting Quality: Illegal logging data may have reporting biases.

Spatial Context Missing: Model lacks precise geographic coordinates.

Time Trend: The observed decline may be influenced by changes in sampling.

9. Recommendations

9.1 Immediate Actions

Action	Priority	Rationale
Deploy Model	High	Model is production-ready with 99.5% accuracy

Action	Priority	Rationale
Create Dashboard	High	Use interactive visualizations for monitoring
Validate Time Trend	Medium	Confirm the 12% decline with consistent sampling
Improve Data Collection	Medium	Add spatial and temporal context

9.2 Future Work

Causal Analysis: Implement Difference-in-Differences or Instrumental Variables

Geospatial Modeling: Integrate satellite imagery and geographic coordinates

Longitudinal Study: Track the same locations over time

Predictive Scenarios: Simulate policy impacts (e.g., fire bans, logging enforcement)

10. Project Deliverables

10.1 Code Artifacts

File	Description
01_exploration_missing_data.ipynb	Data quality assessment
02_feature_engineering.ipynb	Feature creation
02.5_more_feature_engineering.ipynb	Final preprocessing
03_visualization&insights.ipynb	EDA & visualizations
04_model.ipynb	Model training & evaluation

10.2 Model Assets (Ready for Deployment)

File	Description
random_forest_model.pkl	Trained model
label_encoder.pkl	Target encoder
feature_importance.csv	Ranked features
cv_search_results.csv	Tuning results
metrics.txt	Performance metrics

10.3 Visualization Assets (19 Visualizations)

#	File	Type	Description
1	01_risk_distribution.png	Static	Risk level counts
2	02_tree_cover_loss_by_region.png	Static	Regional loss
3	03_tree_cover_loss_by_risk_boxplot.png	Static	Loss by risk
4	04_tree_cover_loss_by_forest_type.png	Static	Loss by forest type
5	05_tree_cover_loss_by_illegal_logging.png	Static	Illegal logging comparison
6	06_average_tree_cover_loss_over_time.png	Static	Time trend
7	07_loss_rate_by_risk.png	Static	Loss rate by risk
8	08_fire_incidents_by_region.png	Static	Fire counts

#	File	Type	Description
9	<code>09_interactive_forest_area_vs_loss.html</code>	Interactive	Scatter plot
10	<code>10_interactive_correlation_heatmap.html</code>	Interactive	Correlation matrix
11	<code>feature_importance.png</code>	Static	Feature importance
12	<code>learning_curve.png</code>	Static	Learning curve
13	<code>per_class_metrics.png</code>	Static	Per-class metrics
14	<code>precision_recall_curves.png</code>	Static	PR curves
15	<code>roc_curves.png</code>	Static	ROC curves
16	<code>class_distribution.png</code>	Static	Class distribution
17	<code>confusion_matrix.png</code>	Static	Confusion matrix
18	<code>cv_search_results.png</code>	Static	CV results
19	<code>Regional_overview.png</code>	Static	Pie charts composite

10.4 Data Assets

File	Description
<code>deforestation_cleaned.csv</code>	Cleaned dataset
<code>deforestation_features.csv</code>	Engineered features

File	Description
<code>deforestation_training.csv</code>	ML-ready dataset